

Automated Compliance: A Privacy-Focused Solution for GDPR and DPDPA Adherence

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Abstract—This paper explores the possible automation of Governance, Risk, and Compliance (GRC) concerning various evolving privacy regulations—specifically, the General Data Protection Regulation (GDPR) and the Digital Personal Data Protection Act (DPDPA). With the increasing complexity of regulatory requirements, especially regarding data privacy, there is also a growing need for an automated GRC solution that will help organisations make sure to adhere to all applicable laws and regulations. The proposed compliance checker tool aims at the automation of several aspects of the audit process. This tool will be a starting point for compliance for organisations, more so for small and medium enterprises (SMEs), which are often beset with issues such as financial problems, lack of expertise, and vendor concentration on thirdparty risk management. The evaluation of this tool on a dataset of 50 websites (25 non-compliant and 25 compliant) yielded promising results, achieving an accuracy of 86%, a recall rate of 92%, and an F1 score of 86.79%. This work has shown that automation of such processes can significantly, if not completely, reduce compliance errors, lower operational costs, and speed up the proceeding audits and assessments, thereby making automation a strong enabler for organisations in the complex regulatory environment to improve the level of compliance for better handling of customer privacy.

Keywords: GRC Automation, GDPR, DPDPA, Governance, Risk Management, Compliance, Auditing, Data Privacy

I. INTRODUCTION

Governance, Risk, and Compliance (GRC) management has been incorporated into business operations more consistently with various sectors, such as finance or health care. The companies further face increasing pressure in coping with the risks and the compliance regulations as the regulatory landscape tends to get richer. In a report published by KMPG, 75% of organisations interviewed agreed that changing regulations evolved to be the most difficult GRC challenge, which therefore calls for constant redesign of policies, processes and controls [1]. Due to such complexity and lack of proper structures, the manual processes of GRC face numerous shortcomings because of inefficiency, higher chances of mistakes and delays in compliance reporting.

These challenges seem to be even starker for small and medium enterprises (SMEs). Studies by the Cyber Peace Foundation indicate that SMEs are especially vulnerable to cyberattacks, and many find it hard to recover if they have been breached, with substantial percentages going out of business due to such occurrences. This, in turn, makes compliance with provisions such as the General Data Protection Regulation (GDPR) and Digital Personal Data Protection Act (DPDPA) not just instructive adherence but survival ones. SMEs face a shortage of resources and capacities to address such complicated regulative conditions.

Introduced in 2018, the GDPR is one of the most comprehensive regulations in the world, reaching out to businesses processing the personal data of citizens of the European Union, and it requires stern guidelines for collection, consent, and protection of data with penalty fines up to 20 million Euros or 4% of global annual turnover, whichever is greater [2]. India introduced the DPDPA in 2023 to protect the personal data of Indian citizens [3]. DPDPA, in its infancy of being practised, seems to have some of the toughest guidelines concerning data collection and use. A recent example of a penalty was the imposition of as much as EUR 1.2 billion on Meta as a result of violations against data transfer guidelines under GDPR. Huge financial liabilities for non-compliance exert extreme pressure on organisations to ensure that privacy practices are deemed robust and transparent.

Automation in GRC is no longer a luxury but a necessity for organisations seeking to stay competitive and compliant. The global Enterprise GRC (eGRC) market, valued at \$54.61 billion in 2023, is expected to grow at a CAGR of 13.8% through 2030, with automation playing a critical role in this expansion [4].

Organisation of the paper: Section 2 discusses the existing tools in the market. Section 3 outlines the methodology. Section 4 presents an experimental analysis of the proposed tool, including the evaluation criteria. Section 5 discusses the results. Section 6 and Section 7 look into the advantages and the applications of the proposed tool, respectively. Section 8 concludes the paper and provides insights into future work.

II. LITERATURE SURVEY

The National Institute of Standards and Technology (NIST) [5] first highlighted the importance of data governance by providing the foundational guidelines for organisations as to how to handle the data securely and responsibly. NIST's work also set the foundation for data management as critical in an increasingly digital world. Following this, many countries have realised that there is a need to have robust data governance in place, and so have introduced laws and regulations that help protect personal data. For example, the European Union has its GDPR, and India has its DPDPA. As a result, data protection has become a global concern, requiring industries and sectors to follow more stringent measures of compliance.

GDPR and DPDPA are two of the strictest data protection laws globally to ensure proper transparency, security, and accountability when personal data is dealt with. Here, while the GDPR regulates processing for citizens from the European Union with drastic penalties for violation, a similar stand has been maintained by the DPDPA towards citizens of India with further enforcement in the future. Both regulations rely on user consent, notification in case of a breach, and the rights of the individual to control their personal information.

Li, He, Yu, Lu, and He, Wu (2019) scrutinised the international impact of GDPR [6]. Here, it is emphasised how such complexity about compliance increases the operational stress of international business, especially tech and cybersecurity, among others such as AI and blockchain. The fact that requirements for data protection, including the rights of access, erase, and portability, bring significant restructuring demands on U.S. and Chinese firms operating within the territory of the EU. Compliance is framed not only as an adherence to the regulation but also as a competitive advantage in consumer trust and data security. GDPR Advisor [7] highlights the conflict between AI's data demands and GDPR's principles of data minimisation and purpose limitation. While AI systems benefit from large datasets to improve accuracy, excessive data collection or use beyond its specified purpose can lead to compliance issues.

The developments of technology, which keeps on improving and the emphasis on data protection and regulation have compelled various organisations to embrace the Governance, Risk and Compliance (GRC) tools as part of the solution to the matter. GRC tools are aimed at assisting organisations in fulfilling legal requirements such as the GDPR, ePrivacy Regulation, and even new ones like the Indian DPDPA. Therefore, these tools minimise the risk of monetary fines and penalties related to non-compliance by automating data management processes, management of data subjects' consents, performing audits and other risk management activities. Some of the most popular tools in this category include OneTrust, Cookiebot, 2GDPR,

Probely, AuditBoard, Vanta and others. All these tools are focused on different areas of compliance management, giving the organisation the ability to satisfy different requirements from various regulations. Table I provides a summary of this market analysis, comparing the tools based on their strengths and limitations, along with how the proposed solution bridges these gaps where applicable.

Table I: Gap Analysis of Existing Tools

Name	Features	Gap Identified
OneTrust	Comprehensive GRC tool for GDPR, CCPA, LGPD; Data mapping, cookie compliance, privacy impact assessments [8].	Complex configuration, high price, unsuitable for SMEs [8].
Cookiebot	Automates cookie compliance; scans websites for cookies and manages user consent [9].	Limited to cookies, lacks advanced GRC features such as audit management [9].
2GDPR	Affordable, userfriendly, focuses on cookie compliance	Limited to cookie management, lacks risk rating and audit
	for very small enterprises [10].	management features [10].
Probely	API penetration testing; helps companies comply with data protection (Article 32 of GDPR) [11].	Focuses on API security, lacks comprehensive GRC compliance management features [11].
AuditBoard	Comprehensive audit, risk, and compliance management for large organizations [12].	High cost, complex to use, not suitable for small teams or SMEs [12].
Vanta	Automates compliance with SOC 2, ISO 27001; live updates, checklists for ongoing compliance [13].	Checklists are rigid and insufficient for larger firms requiring in-depth GRC management [13].

Despite the wide range of services provided by platforms such as OneTrust, Cookiebot, and AuditBoard, there are still several issues that are common in GRC automation. First, most of the tools are narrowband and focused on certain aspects, such as cookie compliance or API security, without offering an integrated solution for the management of multiple compliance regimes. This can lead to a situation where a company must rely on several isolated tools, making the whole process more expensive and cumbersome. On top of that, solutions offered with the likes of OneTrust are featureheavy and tend to intimidate smaller outfits in terms of their use and even price. Another issue that keeps recurring is that of scaling up – solutions such as 2GDPR and Cookiebot are usually affordable and easy to use, but they are not very friendly for large-scale businesses. Furthermore, there is still the challenge of most of these tools lacking sophisticated AIbased predictive abilities that can be able to foresee compliance risks and streamline decision-making. The challenges call for the need for a better solution that is more versatile and can help

handle the compliance requirements of different businesses in different sectors.

III. METHODOLOGY

The proposed compliance checker tool aims to aid organisations with a tailored and specific compliance audit process by providing a list of questions appropriate to their choice of regulations like GDPR or DPDPA that need to be addressed. To ensure that certain quantitative regulations are followed, the tool also offers a script that can be executed on the user's machine to facilitate further validation, followed by the generation of an automated report that examines the noncompliance areas and includes suggestions for correcting these gaps, depending on the answers and the automated evaluation. Besides, the tool also helps indicate the level of compliance and provides a spider chart, which scores the organisation in different compliance categories, as illustrated in Fig. 1. The tool identifies various pillars of compliance requirements, including Processors, Data Transfers, Accountability, Processing Principles, Consent & Notices, Data Subject Rights, Records of Processing, Data Protection Impact Assessment, Privacy by Design & Default, and Breach Management.

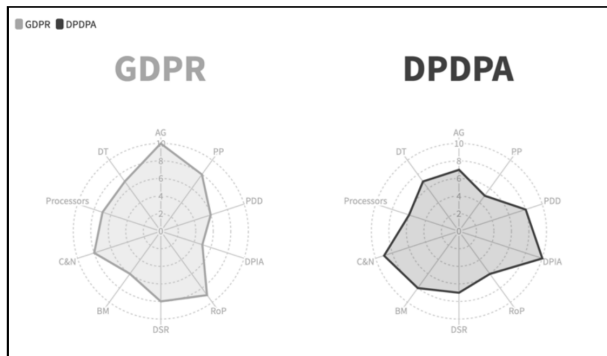


Fig. 1: Spider Chart Generated in the Report

The automation component of the regulatory compliance checker tool currently consists of three modules: Extraction Module, Transparency Module, and Tracking Detection Module. The first two run in sequence to form a pipeline that starts with extracting data moves on to assess for transparency, and then runs the tracking detection module when the components require additional tracking assessment. This ensures full checks towards compliance and incites awareness about tracking.

A. Extraction Module

The Extraction Module is intended to extract cookierelated disclaimers or policies from websites. It makes use of two methods that enable it to accommodate different website implementations:

- *Static Extraction:* It uses BeautifulSoup to parse the HTML content directly. This is best suited for sites that load their content statically.

- *Dynamic Extraction:* Uses Selenium if the content is loaded dynamically using JavaScript. In that scenario, Selenium will ensure to capture the dynamic elements.

Fig. 2 shows the pseudocode for the module that provides the foundational input for the transparency checks by extracting essential compliance data related to cookies and tracking from the website.

```
function extract_cookie_disclaimer(URL,
use_selenium=False):

    if use_selenium is False:

        # Try extracting using static parsing
        (BeautifulSoup)

        result_bs4 =
        extract_cookie_disclaimer_bs4(URL)

        if result_bs4 contains "cookie banner"
        or "disclaimer":

            return result_bs4

        # If static extraction fails or Selenium is
        preferred:

        if use_selenium is True or bs4 fails:

            result_selenium =
            extract_cookie_disclaimer_selenium(url)

            return result_selenium if
            result_selenium contains "cookie banner"

            return "No cookie information found"
```

Fig. 2: Extraction Module Pseudocode

B. Transparency Module

The Transparency Module then calculates a transparency score in consideration of the following factors:

- *Clarity:* readability metric on how easily the text can be understood.
- *Directness:* a basic language that is uncertain - for the direct purpose of communicating.
- *Tracking Mentions:* Presence of data tracking or sharing.
- *Explicitness:* The module favors short and straightforward sentences and penalizes ones that are too long.
- *Stop word Ratio:* Has a high percentage of stop words, potentially lowering the transparency.

Fig. 3 shows the module pseudocode, which generates a transparency score for the disclaimer or policy, which will determine if it complies with regulations on transparency. Suppose the score reveals less than a certain threshold, which might imply non-compliance.

```
function calculate_transparency_score(text):
    readability_score =
    calculate_readability(text)

    clarity_score =
    normalize_score(readability_score)

    hedge_word_count = count_hedge_words(text)

    directness_score = 1 - (hedge_word_count /
    total_words)

    tracking_score = 1 if text mentions
    tracking_keywords else 0

    sentence_length =
    calculate_avg_sentence_length(text)

    explicitness_score = 1 if sentence_length <
    threshold else reduce_score(sentence_length)

    stopword_count = count_stopwords(text)

    stopword_ratio = stopword_count /
    total_words

    stopword_score = 1 - stopword_ratio

    return weighted_combination(clarity_score,
    directness_score, tracking_score,
    explicitness_score, stopword_score)
```

Fig. 3: Transparency Module Pseudocode

C. Tracking Detection Module

The Tracking module inspects for the presence of tracking methods other than cookies. These include:

- *LocalStorage*: This module scans through the browser's local and session storages.
- *Service Workers*: It attempts to identify active service workers, which may cache your activity and track you.
- *Web Beacons and Canvas Fingerprinting*: Pick up hidden image tags and canvas fingerprints.
- *WebRTC Leaks*: Searches for any IP leaks through WebRTC.

Fig. 4 shows the module pseudocode provides more information on the tracking methods carried out on the website to ascertain if what is being carried out is by the tracking regulation. Based on the findings, behaviour that does not comply with the requirements can be flagged for review or rectification.

These modules are integrated into a sequential pipeline that automates the compliance auditing process, as shown in Fig. 5 and Fig. 6. This workflow initially assesses the clarity and transparency of the website's cookie policies. If the transparency score is insufficient or the required data is not found, it proceeds to evaluate the website's tracking mechanisms to ensure compliance with tracking regulations.

```
function detect_tracking(url):
    start_proxy_capture()

    load_website_in_browser(url)

    cookies = detect_cookies(driver)

    local_storage = detect_local_storage(driver)

    session_storage =
    detect_session_storage(driver)

    indexed_db = detect_indexed_db(driver)

    service_workers =
    detect_service_workers(driver)

    web_beacons =
    detect_web_beacons(html_content)

    canvas_fingerprint =
    detect_canvas_fingerprinting(driver)

    etags = detect_etags(url)

    webrtc_leaks = detect_webrtc_leaks(proxy)

    user_agent = detect_user_agent(driver)

    ip_leaks = detect_ip_logging(proxy)

    stop_proxy_capture()

    return
    aggregate_tracking_information(cookies,
    local_storage, service_workers, web_beacons, etc.)
```

Fig. 4: Tracking Module Pseudocode

```
function compliance_audit(url):
    cookie_disclaimer =
    extract_cookie_disclaimer(url, use_selenium=True)

    if cookie_disclaimer contains valid
    information:
        transparency_score =
        calculate_transparency_score(cookie_disclaimer)
        if transparency_score > threshold:
            return "Compliant based on
            Transparency."
        else:
            return "Non-compliant based on
            Transparency."
    else:
        tracking_results =
        detect_tracking(url)
        if tracking_results meet compliance:
            return "Compliant based on
            Tracking."
        else:
            return "Non-compliant based on
            Tracking."
```

Fig. 5: Integrated Automation Workflow

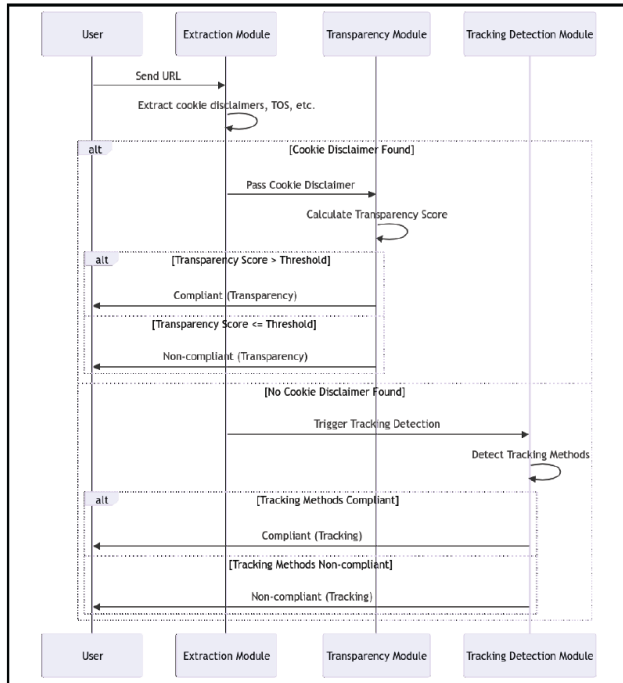


Fig. 6: Sequence Diagram for Integrated Automation Workflow

IV. EXPERIMENTAL ANALYSIS

A. Evaluation Methodology

To evaluate the proposed GDPR compliance tool, a dataset of 50 websites was selected, evenly split into two categories:

- *Non-Compliant Websites (25)*: Identified through Data Protection Authority (DPA) records, historical versions from the Internet Archive were used to capture noncompliance states. Sourced from DPA records, covering infractions like consent mechanism flaws and data security lapses. Wayback Machine archives provided snapshots from violation timelines.
- *Compliant Websites (25)*: Chosen based on certifications (ISO 27001, 27701, etc.) and reputations for strong GDPR practices. Included organizations renowned for GDPR compliance, representing sectors such as finance, healthcare, and e-commerce.

Data was collected using automated tools focused on GDPR-relevant features:

- *Automated Crawling*: Web crawlers gathered data from each website's privacy policies, consent banners, and data collection forms.
- *Data Processing*: We parsed the data to extract key features like consent mechanisms and privacy notices, then deployed the compliance tool on each website to log analysis results.

B. Evaluation Metrics

To measure the tool's effectiveness, standard metrics defining compliance detection accuracy were used:

- *True Positives (TP)*: Non-compliant sites are accurately identified as non-compliant.
- *True Negatives (TN)*: Compliant sites are accurately recognised as compliant.
- *False Positives (FP)*: Compliant sites incorrectly flagged as non-compliant.
- *False Negatives (FN)*: Non-compliant sites mistakenly classified as compliant.
- Additionally, Accuracy, Precision, Recall, Specificity, and F1 Score were calculated to evaluate tool performance.

V. RESULTS AND DISCUSSION

Table II shows the confusion matrix derived from analyzing the 50 websites for GDPR compliance.

Table II: Confusion Matrix for GDPR Compliance

	Actual NonCompliant	Actual Compliant	Total
Identified Non-Complaint	23 (TP)	5 (FP)	28
Identified Compliant	2 (FN)	20 (TN)	22
Total	25	25	50

Table III shows the metrics derived from the above confusion matrix.

Table III: Evaluation Metrics for GDPR Compliance

Metric	Score
Accuracy	86%
Precision	82.14%
Recall (Sensitivity)	92%
Specificity	80%
F1 Score	86.79%

The results outline the tool's strengths and weaknesses. False positives arose due to overly stringent compliance checks. For example, immediate consent banners and strict formats for privacy policies were some of the requirements. It would, therefore, be necessary to fine-tune these evaluations to minimise unnecessary alerts on compliant websites. False negatives largely occurred from the fact that the tool only performed frontend analysis, thus failing to pick on backend compliance issues that are essential for total detection of GDPR compliance. The outcomes indicate a necessity for the inclusion of backend compliance features. In addition, dataset size and archived snapshots of the website could be an important factor for this reason, as archived sites hardly possess the backend functionality along with dynamic content. Results must be compared with manual audits to get further validation and may perhaps highlight the requirement to refine.

VI. ADVANTAGES OF PROPOSED SYSTEM

Automated GRC provides real-time monitoring of compliance and risk across the enterprise to enable real-time corrective measures. Compliance status can be tracked continuously, so companies are always ready for an audit.

The same is true for possible risks; it really will provide stakeholders with clearer views to allow for speedier decision-making processes.

With the complexity of the regulatory and risk management issues facing a growing organisation, the actual GRC platforms should scale with them. Thus, there must be scalability that caters to different needs - large enterprises and small to medium-sized enterprises. However, SMBs and SMEs have adopted automated GRC solutions at a faster rate. For instance, the market for SME-focused GRC tools is expected to grow at the highest CAGR from 2020 to 2030 [4].

The traditional management of GRC involves a lot of manual effort, time-consuming and susceptible to human error. Industry analysts estimate that automation of GRC can slice operational costs by 30-40% [4]. Automation makes compliance workflows smoother and makes it easier to track regulatory changes while reducing administrative burden. It, therefore, becomes more feasible for organisations to sustain compliance status in real-time without requisite resources committed to mundane tasks. The automated GRC solutions are also scalable to meet the needs of small and large businesses in managing increasing regulatory demands without significantly increasing personnel costs.

Advanced GRC has robust analytics and transforms compliance data into actionable insights. Automation in data collection and analysis supports trend identification, pattern discovery, and risk predictions and lets an organisation properly plan to mitigate those risks. This data-driven approach aligns the risk management strategy with the broader goals of the organisation.

VII. APPLICATIONS

A. Information Technology (IT) and Telecommunications

The IT and telecommunications sectors handle enormous amounts of personal data, putting them under strict law and regulatory scans, especially now that there are laws in place in Europe, GDPR and an emerging law in India, DPDPA. In the case of GDPR, the law has been established since 2018. DPDPA, though it was a law made in 2023, is expected to be put in place in a phased manner [3].

Under the GDPR regulations, IT and telecom firms will have to make sure personal data is processed lawfully and transparently. These firms can monitor data usage, manage consent, and respond to breaches in real time through

automated GRC tools. Non-compliance attracts high fines equivalent to up to 4% of global revenue; thus, tracking and auditing of data processing become indispensable with automation.

India's DPDPA, at large, mirrors GDPR, but it still has much more that is in the infancy stage, and the business is currently making requisite arrangements in setting up consent mechanisms, data breach notification systems, and risk management frameworks. Since DPDPA tools are not developed yet, most organisations have taken global compliance tools such as those developed based on GDPR and adapted those to be ready.

Both industries are highly exposed to cyberattacks that require strong GRC automation for real-time monitoring of vulnerabilities as well as overall adherence to international cybersecurity standards, such as ISO/IEC 27001. An automated system allows real-time tracking of data security, and it produces alerts if there might be breaches, thus saving time in response.

B. Banking, Financial Services, and Insurance (BFSI)

The financial services sector is one of the most highly regulated sectors, covering all kinds of laws varying from Basel III and MiFID II and Dodd-Frank to anti-money laundering compliance. In such industries, the GRC processes become critical in the management of risks and compliance continuously.

Banks and financial institutions require constant evaluations of credit and operational risks. Financial organisations can continuously monitor credit risks and employ AI to detect early warning signs of loan defaults through GRC automation. It minimises the risks of nonperforming assets and helps institutions comply with Basel III guidelines. Financial institutions must prepare regular reports for regulatory authorities like the US Securities and Exchange Commission (SEC) and Financial Conduct Authority (FCA), for which a reporting tool is necessary. GRC automation will compile such reports from various data sources, ensuring accuracy and timeliness. Automating processes involved in regulatory reporting reduces human error and increases efficiency in compliance.

C. Healthcare

Severe privacy and security regulations affect the US healthcare industry through the Health Insurance Portability and Accountability Act (HIPAA), while those in Europe face GDPR. Such regulations determine how a healthcare provider handles sensitive information about patients, and this makes GRC automation an integral part of healthcare activities.

HIPAA and GDPR ensure that Personal Health Information (PHI) is covered by the health providers. GRC automation is permanent in monitoring access to the

data and reporting unauthorised attempts of access. They give audit compliance reports as well. The systems ensure security from cases of data breaches since they ensure that any case concerning PHI is given all its regulatory requirements. Apart from having a compliance focus, GRC tools bring in clinical risk management in a few places, like adverse event tracking and medication errors. The automated systems help health organisations identify the risks that may have been possible, hence improving patient outcomes and reducing liability.

VIII. CONCLUSION AND FUTURE SCOPE

With increasing complexity in regulatory requirements, automating GRC processes is no longer optional but a necessity. As discussed in this paper, automation of compliance tasks allows for streamlined operations, lower human error, cost reductions, and time savings. By using tools like web scraping, natural language processing, and tracking detection, the proposed tool helps organisations stay compliant with regulations such as GDPR and DPDPA while providing real-time monitoring and transparency.

Automation in GRC has contributed significantly to the financial, health, and IT industries that operate with the key elements of risk management and compliance daily. Extraction of cookie policies for automating transparency evaluation and detection of tracking would improve the ability of an organisation to safeguard user data, fulfilling increased regulatory needs. An increasingly regulated world faces this problem, and automation of GRC is one of the effective, scalable solutions for these problems.

PECB Insights [14] asserts that AI-based systems introduce unique challenges in ensuring compliance with the GDPR, especially concerning transparency and accountability. The models used by AI are “black boxes,” making it difficult for companies to explain the decisions made by an automated process to the data subject as required by Article 22 of the GDPR. PECB promotes the adoption of explainable AI methods, such as rule-based models, that help companies maintain compliance through insight into AI decisions. This approach deals with the demand for transparency set by GDPR that affected people understand how choices that affect them are made.

OXYGY Consulting’s Friedemann Lutz and Lennart Schüßler [15] point out user-centric design for the optimisation of the GDPR process, such as DPIA. The classic processes of compliance with general data protection regulation are essentially inefficient and resource-intensive. By incorporating design thinking in compliance processes, the company can attain streamlined compliance workflow, increase efficiency, and reduce resource usage. This encourages more ownership of GDPR

practices within business units, making GDPR from a standalone task to a core business process.

Looking ahead, promising opportunities for better techniques can improve compliance with the GDPR. Entropy analysis can be applied to assess encryption strength to ensure that sensitive data remains securely protected. As an added check, packet inspection helps data move safely across the web by catching problems related to outdated or weak encryption. Companies could identify weaknesses in real time and repair these before they could cause trouble by automating the steps. API testing would also ensure that requests for personal data, say, the access or deletion of information, are managed and followed with GDPR compliance.

This advances the ability of companies to set up a comprehensive whole system for real-time checking of compliance, greatly reduces hands-on work involved with it, and increases data safety. As AI and Machine Learning improve, these solutions will not only make an organisation rule-abiding, but they will also start to see and prevent their risks before they materialise.

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